Integration

ANSWER KEY

Section 1: Standard Integrals (Trig, Exp & Rational)

1 Find $\int \frac{2}{3x-1}\,dx$.

a. $\frac{2}{3}\ln|3x-1|+C$ ✓

b. $2\ln|3x-1|+C$

c. $\frac{1}{3}\ln|3x-1|+C$

d. $-\frac{2}{(3x-1)^2}+C$

$\int \frac{1}{ax+b}dx = \frac{1}{a}\ln|ax+b|+C$. Multiplying by 2 gives $\frac{2}{3}\ln|3x-1|+C$.

2 Find $\int \cos^2(x)\,dx$.

a. $\frac{x}{2}+\frac{\sin(2x)}{4}+C$ ✓

b. $\frac{\sin^3(x)}{3}+C$

c. $\frac{x}{2}-\frac{\sin(2x)}{4}+C$

d. $\sin(2x)+C$

Use identity $\cos^2 x = \frac{1+\cos(2x)}{2}$. $\int \frac{1+\cos(2x)}{2}dx = \frac{x}{2}+\frac{\sin(2x)}{4}+C$.

3 Evaluate $\int_0^{\pi/4} \tan(x)\,dx$.

a. $\frac{1}{2}\ln 2$ ✓

b. $\ln 2$

c. 1

d. $\frac{\sqrt{2}}{2}$

$\int \tan x dx = [-\ln|\cos x|]_0^{\pi/4} = -\ln(\sqrt{2}/2)+0 = -\ln(2^{-1/2}) = \frac{1}{2}\ln 2$.

Section 2: Integration by Substitution

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4 Evaluate $\int x\sqrt{x^2+5}\,dx$.

a. $\frac{1}{3}(x^2+5)^{3/2}+C$ ✓

b. $\frac{1}{2}(x^2+5)^{3/2}+C$

c. $\frac{2}{3}(x^2+5)^{3/2}+C$

d. $(x^2+5)^{3/2}+C$

Let $u=x^2+5$, $du=2xdx$. $\int \frac{1}{2}u^{1/2}du=\frac{1}{2}\cdot\frac{2}{3}u^{3/2}+C=\frac{1}{3}(x^2+5)^{3/2}+C$.

5 Evaluate $\int_0^1 \frac{x}{x^2+1}\,dx$.

a. $\frac{1}{2}\ln 2$ ✓

b. $\ln 2$

c. $\frac{\pi}{4}$

d. 1

Let $u=x^2+1$, $du=2xdx$. Integral from $u=1$ to $u=2$: $\int_1^2 \frac{1}{2u}du=[\frac{1}{2}\ln u]_1^2=\frac{1}{2}\ln 2$.

6 Use the substitution $x=2\sin\theta$ to find $\int\sqrt{4-x^2}\,dx$.

a. $2\theta+\sin(2\theta)+C$ ✓

b. $4\theta+2\sin(2\theta)+C$

c. $2\arcsin(x/2)+C$

d. $\frac{1}{2}(4-x^2)^{3/2}+C$

$dx=2\cos\theta d\theta$, $\sqrt{4-x^2}=2\cos\theta$. $\int 4\cos^2\theta d\theta=\int 2(1+\cos 2\theta)d\theta=2\theta+\sin(2\theta)+C$.

Section 3: Integration by Parts

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7 Find $\int xe^{3x}\,dx$.

a. $\frac{xe^{3x}}{3} - \frac{e^{3x}}{9} + C$ ✓

b. $\frac{xe^{3x}}{3} - \frac{e^{3x}}{3} + C$

c. $xe^{3x} - e^{3x} + C$

d. $\frac{x^2e^{3x}}{6} + C$

Let $u = x, dv = e^{3x}dx \Rightarrow du = dx, v = \frac{1}{3}e^{3x}$. $uv - \int v du = \frac{1}{3}xe^{3x} - \int \frac{1}{3}e^{3x}dx = \frac{xe^{3x}}{3} - \frac{e^{3x}}{9} + C$.

8 Evaluate $\int_1^e x\ln x\,dx$.

a. $\frac{e^2+1}{4}$ ✓

b. $\frac{e^2-1}{4}$

c. $\frac{e^2}{2}$

d. $\frac{e^2-1}{2}$

$u = \ln x, dv = xdx \Rightarrow du = dx/x, v = x^2/2$. $[\frac{x^2}{2}\ln x]_1^e - \int_1^e \frac{x}{2}dx = \frac{e^2}{2} - [\frac{x^2}{4}]_1^e = \frac{e^2}{2} - (\frac{e^2}{4} - \frac{1}{4}) = \frac{e^2+1}{4}$.

9 Evaluate $\int_0^\pi x\sin x\,dx$.

a. π ✓

b. 2π

c. 0

d. 2

$u = x. dv = \sin x dx \Rightarrow du = dx. v = -\cos x.$
$[-x\cos x]_0^\pi - \int_0^\pi (-\cos x)dx = -\pi(-1) - 0 + [\sin x]_0^\pi = \pi + 0 = \pi.$

Section 4: Partial Fractions Integration & Trapezium Rule

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10 Find $\int \frac{5x-2}{(x-2)(x+1)}\,dx$.

a. $\frac{8}{3}\ln|x-2|+\frac{7}{3}\ln|x+1|+C$ ✓

b. $3\ln|x-2|+2\ln|x+1|+C$

c. $8\ln|x-2|+7\ln|x+1|+C$

d. $\ln|(x-2)(x+1)|+C$

Partial fractions: $\frac{A}{x-2}+\frac{B}{x+1}$. For A: $(5(2)-2)/(2+1)=8/3$. For B: $(5(-1)-2)/(-1-2)=-7/(-3)=7/3$. Integrating gives $\frac{8}{3}\ln|x-2|+\frac{7}{3}\ln|x+1|+C$.

11 Use the trapezium rule with 4 intervals (5 ordinates) to approximate $\int_1^3 \frac{1}{x}\,dx$.

a. 1.117 ✓

b. 1.099

c. 1.150

d. 1.050

$h=(3-1)/4=0.5$. $x=1, 1.5, 2, 2.5, 3$. $y=1, 2/3, 1/2, 2/5, 1/3$. Area $\approx \frac{0.5}{2}[1+1/3+2(2/3+1/2+2/5)]=0.25[4/3+2(47/30)]=0.25[40/30+94/30]=0.25(134/30)=134/120$.

12 Is the trapezium rule approximation in the previous question an overestimate or underestimate, and why?

a. **Overestimate, because the curve $y=1/x$ is concave up** ✓

b. Underestimate, because $y=1/x$ is decreasing

c. Overestimate, because the intervals are too wide

d. Underestimate, because the chords lie below the curve

Since $d^2y/dx^2=2/x^3>0$ for $x>0$, the curve is convex (concave up). The straight line chords lie above the curve, producing an overestimate.

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